Customer Behavior Analysis

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About the Dataset

Type: E-commerce Dataset
Nature: Tabular
Total Records : 5,000 customers
Columns : 17
Data Mix:

Numerical → Age, Purchase Amount, Previous Purchases
Categorical → Category, Payment Method, Season

This dataset is used for customer behaviour analysis and business decision-making in the e-commerce domain.

Why This Data Is Important

This data helps businesses:

- Identify **high-value customers**
  Understand **purchase patterns**
- Optimize **discount strategies**
- Improve **marketing campaigns**
- Increase **customer lifetime value (CLV)**

Business Problem

How can an E-Commerce company understand customer purchasing behavior to increase revenue, improve customer retention, and optimize marketing strategies?

Breaking Down the Problem

Low Customer Retention

Customers are not purchasing frequently

Need to understand:

- Who are frequent buyers
- Who are one-time buyers?

Ineffective Marketing Campaigns

Discounts and promo codes may not be effective

Questions:

- Dodiscounts increase purchases?
- Which customers respond to promo codes?

Product Demand Understanding

- Which category performs best?
- Which products generate the highest revenue?

Payment Behavior Insights

- Which payment methods are preferred
- Are digital payments increasing?

Seasonal Demand Variation

- Does season impact sales?
- Which season drives maximum purchases?

Customer Segmentation Problem

Need to identify:

- High-value customers
- Frequent buyers
- Discount-sensitive customers

Data Dictionary

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male/Female/Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Size	Product size (S, M, L, etc.)
Color	Product color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No), Subscription Program
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA- (Exploratory Data Analysis using Python)

Data Loading

What you did:

- Imported **pandas**
- Loaded dataset using:

```
# import dataset
df=pd.read_csv("customer_shopping_behavior.csv")
```

Why did you do it:

- To bring raw data into a structured DataFrame
- Enables further analysis and transformations

```
#observing dataset
df.head(10)
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequ Purch
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash	W
1	521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	20.0	Debit Card	Quar
2	3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	18.0	Cash	Mo
3	1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	25.0	Cash	Ann
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	L	Yellow	Spring	3.6	No	Store Pickup	No	17.0	Credit Card	W
5	3583	23	Female	Shorts	Clothing	41.0	California	S	Beige	Summer	3.4	No	2-Day Shipping	No	15.0	PayPal	Ann
6	1135	62	Male	T-shirt	Clothing	81.0	Wisconsin	M	Brown	Summer	NaN	No	Express	Yes	2.0	Venmo	Ev Mc
7	1411	61	Male	Hat	Accessories	93.0	Tennessee	M	Purple	Summer	4.0	No	Store Pickup	Yes	18.0	Debit Card	Ann
8	113	37	Male	Gloves	Accessories	65.0	Florida	M	Red	Spring	4.3	Yes	Free Shipping	Yes	28.0	PayPal	Bi-W
9	166	42	Male	Handbag	Accessories	60.0	Kentucky	S	Yellow	Winter	2.8	Yes	Store Pickup	Yes	7.0	Cash	Mo

Initial Data Exploration

What you did:

Viewed data using:

- df.head()
- df.tail()

Checked dataset summary:

- df.describe(include='all')
- df.info()

Checked missing values:

- df.isnull().sum()

Why did you do it?

To understand:

- Data structure
- Data types
- Missing values

This step helps identify **data quality issues early**

Data Consistency Check (Category Correction)

What you did:

Checked unique values in:

- Item Purchased
- Category

Found inconsistencies between **Item Purchased and Category**

Created a **mapping dictionary**

Updated category

Why did you do it?

Categories were inconsistent or incorrect

Ensures:

- Accurate grouping
- Correct business insights

Handling Missing Values (Size Column)

What you did:

Handled missing values based on category:

- Electronics → NA
- Accessories → NA
- Clothing → Filled with **mode**
- Footwear → Free Size.

Why did you do it?

Different categories require **different business logic**

Example:

- Electronics don't have size → NA
- Clothing → size is important → use mode

This shows **domain understanding**

Handling Missing Values (Review Rating)

What you did:

Checked :

Mean, Median, Mode

Filled missing values using:

Rounded values

Why did you do it :

Insteadof the global average, the **product-level mean is used**

Thiskeeps ratings **realistic and meaningful**

This is an **advanced imputation technique**

Handling Missing Values (Purchase Amount)

What you did : Replaced with the mean value.

Why did you do it ?

- Maintains **product-level pricing consistency**
- Avoidsdistortion in revenue analysis

Handling Missing Values (Previous Purchases)

What you did: Replaced with the 0 (zero) value.

Why did you do it:

Missing value simply **no previous purchase**

Important for:

- Customer segmentation
- Loyalty analysis

Duplicate Handling

What you did:

- Checked duplicates:
- Identified duplicates using Customer ID
- Removed duplicates:

Why did you do it?

Ensures:

- Each customer appears only once
- No double-counting in analysis

Critical for **accurate reporting**

Column Standardization

What you did:

Converted column names :

- df.columns = df.columns.str.replace(" ","_")
- df.columns = df.columns.str.lower()

```
# renaming
df.columns=df.columns.str.replace(" ","_")
df.columns=df.columns.str.lower()
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases'],
      dtype='str')
```

```
# removing usd from column name as it is more in length
df=df.rename(columns={"purchase_amount_(usd)":"purchase_amount"})
```

Why did you do it?

Makes column names:

- Clean
- SQL-friendly
- Easy to use in Power BI

Data Export to SQL Server

What you did :

Installed required libraries:

!pip install mysql-connector-python sqlalchemy
%pip install pymysql

Connected to SQL Server
Exported data

```
# SQL table name you want to create
table_name = "customer_behaviour"

# Load DataFrame into MySQL table
df.to_sql(
    table_name,
    engine,
    if_exists="replace",   # replace table if already exists
    index=False
)

print(
    f"Data successfully loaded into table '{table_name}' "
    f"in MySQL database 'Customer_behaviour_analysis'."
)
```

```
Data successfully loaded into table 'customer_behaviour' in MySQL database 'Customer_behaviour_analysis'.
```

Why did you do it?

To move cleaned data into **SQL for further analysis**

Enables:

- Query-based insights
- Integration with BI tools.

Data Analysis using SQL Server

1. Which category generates the highest revenue?

category	highest_revenue
Electronics	486367

2. Are discounts actually increasing purchase value?

discount_applied	total_revenue	average_revenue
No	502241.64	182.24
Yes	491940.78	219.22

3. What is the total revenue generated by male and female customers?

gender	highest_revenue
Male	390964.53
Female	352060.29
Other	251157.6

4. **Which customer used a discount but still spent more than the average purchase amount? (Only limited is shown)**

customer_id	purchase_amount	discount_applied
4568	276.3437899543379	Yes
4256	734.4773493975904	Yes
4289	1282.44	Yes
4576	831.3445652173913	Yes
4319	685.2314705882353	Yes
4708	735.773625	Yes
4584	685.2314705882353	Yes
4607	557.24	Yes
4851	685.2314705882353	Yes
4845	692.11	Yes
4558	208.02	Yes
4492	805.502027027027	Yes
4292	805.502027027027	Yes

5. **Which are the top/bottom 5 products with the highest average review rating?**

item_purchased	Avg_ratings
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.79

item_purchased	Avg_ratings
Phone	2.94
Headphones	3.08
Watch	3.1
Bag	3.17
Laptop	3.19

6. **Average purchase: Standard vs Express shipping**

shipping_type	order_placed	avg_purchase	revenue
Express	1199	358.12	429380.98
Standard	1201	340.57	409020.45

7. **Top 5 products with highest discount usage %**

item_purchased	total_number_of_times_sold	Number_of_times_sold_when_disocunt_applied	discount_percent
Laptop	143	74	51.74825
Phone	171	88	51.46199
Headphones	166	85	51.20482
Hat	154	77	50.00000
Watch	157	78	49.68153

8. **Segment customers into new, returning, and loyal based on their total number of previous purchases, and show the count of each segment.**

customer_segment	customer_count
Loyal Customers	3085
Returning Customer	1354
New Customer	561

9. **Do subscribed customers spend more? Compare average speed and total revenue between subscribers and non-subscribers.**

subscription_status	users	avg_revenue	total_revenue
No	3421	165.72	566937.02
Yes	1579	270.58	427245.4

10. What are the top 3 most purchased products within each category ?

category	item_purchased	most_purchased	RNK
Accessories	Jewelry	171	1
Accessories	Bag	164	2
Accessories	Sunglasses	161	3
Accessories	Belt	161	3
Clothing	Shirt	315	1
Clothing	Pants	171	2
Clothing	Blouse	171	2
Electronics	Phone	171	1
Electronics	Headphones	166	2
Electronics	Watch	157	3
Footwear	Shoes	303	1
Footwear	Sandals	160	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

11. **What is the revenue contribution of each age group?**

age_group	total_revenue
51+	399345.32
35-50	266706.7
26-35	177421.93
18-25	150708.48

12. **Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?**

customer_type	subscription_status	customer_count	percents
Normal Buyers	No	658	62.84623
Normal Buyers	Yes	389	37.15377
Repeat Buyers	No	2763	69.89628
Repeat Buyers	Yes	1190	30.10372

Power BI Dashboard

THANK YOU